IT 630 — Midterm Exam

Tuesday, October 13, 2026 • Individual work • 30 points
30 multiple-choice questions • Select one answer, A–E, for each question.
Estimated time: about 30 minutes (1 minute per question). Time available: 50 minutes (full class period).
Coverage: Weeks 1–6, material through Friday, October 9: problem framing and reproducibility, data forms, APIs and scraping, Git, SQL queries and reshaping, keys and joins, window functions, and star schemas. No notes. Scratch paper is provided. All numerical scenarios are hypothetical. Use consistent units when calculating your answers. Questions inside a named scenario share the facts in its scenario box; each question can be answered without relying on an earlier answer of yours.

Name: _____ Date: _____

Part 1 — Problem framing, reproducibility and data forms

1. A team writes: “We will count gym card swipes by hour for Fall 2026 and share the counts with the recreation director.” Using the course's definition of a problem statement, which element is missing?

A. Who will use the result
B. What decision the result will inform
C. Which period of data is in scope
D. Which data source the team will use
E. What the team will measure

2. While preparing gym data, a team finds that the card readers record entries but not exits. The project's question asks how long visits last. According to the course's lifecycle, what happens next?

A. Continue to analysis, since preparation ends once the data is clean
B. Record every visit length as zero and continue the analysis
C. Skip ahead to reporting and mention the gap in a footnote
D. Go back: collect different data or revise the question
E. Add data management as a sixth stage that repairs the gap

3. A sleep study's participant recordings may not leave the university's secure server. Which repository setup still lets an approved researcher reproduce the reported numbers?

A. Commit the recordings after first renaming each file to remove names
B. Leave the recordings out and describe the main findings in the README
C. Declare the work not reproducible and share only the final charts
D. Commit a small random sample of the recordings for testing the code
E. Commit code and environment, plus the data version and access route

4. A slide reports “Average wait: 14 minutes.” Which record lets someone trace that number back to where it came from?

A. The input data version, code commit and parameters
B. The analyst's name and the date the slide was presented
C. A screenshot of the chart that displays the 14-minute average
D. The final report's file size and the software license
E. A note that the number was checked carefully before presenting

5. A CSV of lab specimens has a sample_id column with values such as 0071 and 0408. After loading, the column shows 71 and 408. What explains this, and what prevents it?

A. CSV compresses repeated zeros, so save the file again without compression
B. Leading zeros are invalid in CSV, so renumber the samples starting at 1
C. The IDs are quantities, so 71 and 408 are the correct values to keep
D. CSV declares no types, so the reader guessed integers; read it as text
E. The file was damaged in transfer, so download it again to restore them

Part 2 — Data types, formats and getting data

6. Two files list clinic appointments for two different weeks. Both have the columns appointment_id, patient_id and date, with one row per appointment. No appointment appears in both files. You want one table containing every appointment from both weeks. What should you do?

A. Stack their rows into one table, keeping the same columns
B. Join the files side by side, matching rows on patient_id
C. Group by patient_id, keeping one summary row per patient
D. Pivot the dates into columns, keeping one row per patient
E. Keep only appointments whose IDs appear in both files

7. A 2 GB table is shared as CSV. A colleague proposes switching to Parquet. Which statement about Parquet is accurate?

A. It turns the file into a database server that others can query
B. It is plain text, so it can be edited in any text editor
C. It stores a schema, so column types travel with the data
D. It removes the need to document what one row of the table means
E. It is a zipped copy of the CSV, so it adds no new information

8. You need all records matching your query. An API response body holds 50 records and the field "next": "page=2". Your script saves the 50 records and stops. What is wrong?

A. Nothing; an API sends every matching record in one single response
B. It saved one page; it must keep following next until none remains
C. The status code must have been 429, so the request was refused
D. The records must be requested again in another format, such as CSV
E. The body is malformed, because a record list cannot contain a link

9. One JSON order record has the fields order_id and customer, plus a list of 4 items. You build a table with one row per item. What results?

A. 1 row, with the four items lost
B. 1 row with 4 item columns, one per item
C. 4 rows, with order_id filled in only the first
D. 5 rows: one for the order and one per item
E. 4 rows, with order_id and customer repeated on each

10. You need data from a website that has no API. Before writing any scraping code, what should you check first?

A. How many requests per second the server can handle before it slows down
B. Whether the page's HTML loads in your browser without showing errors
C. Whether the data on the page would produce an attractive chart
D. Its robots.txt and terms of service, to see if collection is allowed
E. Whether a classmate has already scraped the same site this term

Part 3 — Version control, reshaping and SQL basics

11. You commit config.py with a database password in it. In your next commit you delete the password from the file. Is the password out of the repository?

A. No; the earlier commit still contains it
B. Yes; the newest commit replaces every earlier version
C. Yes, once the deletion has been pushed to the remote
D. No, but only because config.py is not in .gitignore
E. Yes; Git keeps only the latest difference for a file

12. In Git, what is the purpose of staging changes before making a commit?

A. To upload all changed files directly to the remote repository
B. To delete older versions from the repository's history
C. To select the changes recorded by the next commit
D. To run the changed code and check that it works
E. To merge every branch into a single version automatically

13. A wide table has one row for each of 5 garden beds and four yield columns, week1 to week4, with every cell filled. You lengthen it into columns bed, week and yield. How many rows result, and what is one row?

A. 5 rows; one bed
B. 4 rows; one week
C. 20 rows; one bed with four week columns
D. 9 rows; one bed or one week
E. 20 rows; one bed in one week

14. A long table of daily step counts has one row per participant per day. P1 and P2 have rows for days 1–3. P3 joined on day 2, so it has rows for days 2 and 3 only. You pivot to one row per participant with one column per day. What should P3's day 1 cell hold?

A. 0, because P3 recorded no steps on that day
B. P3's average of its two other days
C. Blank (NULL), because no measurement exists for that day
D. P3's day 2 value, copied back to day 1
E. Nothing; P3's row is left out of the wide table

15. Which description of SQL matches the course?

A. A language for querying tables, run by a database engine
B. A database product that is sold by a single company
C. A file format for storing tables on disk, like CSV
D. A Python library for drawing charts from table data
E. A spreadsheet program for typing in data by hand

Part 4 — SQL queries

> Scenario: library loans. The table loans has one row per loan, with columns loan_id, member_id, branch, loan_date and days_kept. It holds 12 loans made by 5 members at 3 branches; each branch has at least one loan. Exactly 4 loans have days_kept greater than 14.

16. SELECT loan_id, branch, days_kept FROM loans WHERE days_kept > 14 ORDER BY days_kept DESC; How many rows are returned, and what does one row represent?

A. 12 rows; one row is one loan, now sorted
B. 4 rows; one row is one loan
C. 4 rows; one row is one member with a late loan
D. 1 row; the row holds the total of the long loans
E. 3 rows; one row is one branch

17. SELECT branch, COUNT(*) AS n_loans FROM loans GROUP BY branch; How many rows are returned, and what does one row represent?

A. 3 rows; one row is one branch
B. 12 rows; one loan alongside its branch's count
C. 5 rows; one row is one member
D. 1 row; the total number of loans
E. 3 rows; one row is one loan from each branch

18. Using loans, you need the number of loans per branch per month. Which sequence produces that table?

A. Group by branch, then derive a month column, then count
B. Count all loans, then divide the total evenly across the months
C. Widen the table to one column per month, then filter to one branch
D. Sort by loan_date, then keep the first loan in each month
E. Derive a month column from loan_date, group by branch and month, then count

Part 5 — Keys and joins

> Scenario: tutoring center. sessions has one row per tutoring session (session_id, student_id, minutes): 10 rows. students is meant to hold one row per student (student_id, major). In fact, student S07 appears twice in students, and student S09 does not appear at all. S07 has 3 sessions and S09 has 2; each of the other 5 sessions matches exactly one row in students.

19. In sessions, the column student_id refers to a row in students. What is student_id in sessions?

A. The primary key of the sessions table
B. A foreign key to students
C. A measure to be totaled in reports
D. A duplicate column to delete
E. A derived column computed from minutes

20. S07's two rows in students list different majors, Analytics and Data Science. You join sessions to students and total the minutes by major. What happens to S07's 3 sessions?

A. They are dropped, because S07 has no single major to group them under
B. They are split evenly, with half of each session's minutes going to each major
C. Each session's full minutes count under both majors, double-counting S07's contribution
D. They are counted once, under whichever of S07's majors is listed first
E. The query fails with an error until the duplicate S07 row is removed

21. You switch to a LEFT JOIN from sessions to students. What happens to S09's 2 sessions?

A. They are dropped from the result
B. They take the major of the row listed just above
C. They merge into one row holding their total minutes
D. They stay in the result, with NULL for major
E. The query fails until S09 is added to students

22. In a different project, a join of a 10-row fact table to a lookup table returns exactly 10 rows. A teammate says this proves every fact row found exactly one partner. What is the best response?

A. Correct; matching counts mean every row found exactly one partner
B. Not proven; a duplicate key can add what a missing key removes
C. Correct, provided that the query used an inner join rather than a left join
D. Not proven, because joins always return more rows than their input table
E. Not proven, because row counts tell you nothing about a join

23. Back in the tutoring tables, which check directly detects that a student_id appears more than once in students?

A. Compare COUNT(*) with COUNT(DISTINCT student_id) in the joined result
B. Compare the row count of sessions with the number of students
C. Sort students by major and look at the first row
D. Sum the student_id column and check for a round number
E. Compare COUNT(*) with COUNT(DISTINCT student_id) in students

Part 6 — Window functions and denominators

> Scenario: dorm water use. The table usage has one row per dorm per day (dorm, day, gallons). It covers 3 dorms over the same 5 days, Monday to Friday: 15 rows. The day column is numbered 1–5, with Monday = 1 and Friday = 5.

24. Query 1 adds AVG(gallons) OVER (PARTITION BY dorm) as a column to every row of usage. Query 2 uses GROUP BY dorm with AVG(gallons). How many rows does each return?

A. Query 1: 3; Query 2: 3
B. Query 1: 15; Query 2: 15
C. Query 1: 3; Query 2: 15
D. Query 1: 15; Query 2: 3
E. Query 1: 5; Query 2: 3

25. In LAG(gallons) OVER (PARTITION BY dorm ORDER BY day), what does PARTITION BY dorm do?

A. Sorts the rows alphabetically by dorm name before they are displayed
B. Collapses each dorm's five days into a single summary row
C. Removes any dorm that is missing a reading on one or more days
D. Saves each dorm's rows as a separate new table in the database
E. Keeps comparisons within one dorm, so its first day has no earlier row

26. Suppose the same readings had arrived as one row per day, with columns dorm_A, dorm_B and dorm_C. Before you can write LAG(gallons) OVER (PARTITION BY dorm ORDER BY day), what has to happen?

A. Sort the table by day so the rows are in date order
B. Add a row-number column so that each row has an ID
C. Lengthen it to one row per dorm per day
D. Nothing; LAG can compare the three dorm columns directly
E. Widen it further to one column for each dorm and day

27. You calculate each day's change as gallons minus the previous day's gallons within the same dorm. Each dorm's Monday change is NULL. A teammate replaces those NULLs with 0 before averaging the changes. What is wrong with that?

A. Monday had no earlier day, so 0 invents a comparison
B. Nothing; 0 and NULL mean the same thing in an average
C. AVG cannot run on a column that once held NULL values
D. Each Monday should be given Tuesday's change instead
E. Each NULL should be replaced with that Monday's gallons

Part 7 — Star schemas

> Scenario: music plays. A music app's flat export has one row per play. It is being split into a fact table plays (play_id, song_id, user_id, play_date, minutes_listened) and a dimension table songs (song_id, title, artist, genre).

28. Genre is stored in songs, and minutes_listened is stored in plays. Which steps give total minutes listened by genre?

A. Sum minutes_listened in songs, grouped by genre
B. Group plays by song_id, then sort the result by genre
C. Join songs to itself on genre, then count the rows
D. Join plays to songs, group by genre, sum minutes_listened
E. Group plays by user_id, then sum minutes_listened

29. The flat export contains May plays only and also repeats each user's monthly subscription fee, $12, on every play row. An analyst sums the fee column to report May subscription revenue. What is wrong?

A. Nothing, since each play row carries the correct fee amount
B. It counts each user's one monthly fee once per play
C. The fee should be averaged across all of the users first
D. The fee belongs in the songs table, not in the flat export
E. Fees can be summed correctly once the rows are sorted by user

30. After the split, which check finds plays whose song_id has no matching row in songs?

A. Comparing the export's row count with the row count of plays
B. Comparing COUNT(*) with COUNT(DISTINCT song_id) in songs
C. Left join plays to songs; count rows where songs.song_id is NULL
D. Summing minutes_listened across all of the rows in plays
E. Sorting the songs table by title and scanning it by eye